Di Yang

Williamsburg, VA | +1 757-332-9351 | dyang06@wm.edu
ioet-y.github.io | Google Scholar | GitHub | LinkedIn

Research Interests

Generative models & world models: flow matching, few-step 3D generation, and compact 4D scene states.

Robot learning & planning: VLA manipulation, sim-to-real transfer, and multi-agent planning.

Agents & multimodal reasoning: continual skill acquisition, tool use, and video-language reasoning.

Education

William & Mary | Ph.D. in Data Science Sep 2024 - Expected Dec 2027
Advisor: Dr. Yanhai Xiong; research collaboration with Dr. Haipeng Chen.

National University of Singapore | M.Sc. in Electrical Engineering Jan 2021 - Jan 2022
Advisor: Dr. John S. Y. Ho; research internship with Dr. Xuejiao Zhao.

University of Electronic Science and Technology of China Sep 2016 - Jul 2020
B.Eng. in Microelectronic Science and Engineering.

Publications & Preprints

Peer-reviewed / Accepted

[1] VTV-FM: Flow Matching through Variational Terminal-Velocity Closure.
H. Jiang*, Y. Li*, **Di Yang***, Y. Xiong, H. Chen, Y. He. **NeurIPS 2026** (accepted, poster). [Code]

[2] HyCO: A Hybrid Neural Solver for Combinatorial Optimization.
Y. Li*, **Di Yang***, H. Chen, Y. Xiong. **NeurIPS 2026** (accepted, poster). [Code]

[3] LENS: Learning to Navigate with Active Search for Partially Observable MAPF in Unknown Environments.
Di Yang, Y. Li, Y. Xiong. **Transactions on Machine Learning Research (TMLR), 2026**. [Code]

[4] UP-Bench: A Benchmark for Underwater Path Planning Algorithms.
Di Yang, Y. Xiong. **KDD 2025, Benchmark Track (Oral)**. [Code]

**Equal contribution / co-first author.*

Preprints & Technical Reports

[5] Robot Skill Factory: Towards Recursive Self-Improvement in Humanoid Control.
Di Yang. Technical report, 2026. [Code]

[6] Channel Match: Calibrating Shape Latents for Few-Step 3D Generation.
Di Yang, Z. Li, Y. Xiong, Y. Wang. Under review, 2026. [arXiv]

[7] A Compact Explicit 4D Representation for Dynamic Scenes.
Di Yang, Z. Li, Y. Xiong, Y. Wang. Preprint, 2026. [arXiv]

[8] What Explicit Physics Buys a Manipulation World Model.
Di Yang, M. Liu, J. Chen, Y. Li, Y. Wang, Y. Xiong. Preprint, 2026. [Code]

[9] Ask About the Right Moment: Training-Free Object Identity Reasoning for Video-Language Models.
Di Yang, X. Li, M. Liu, Y. Wang, Y. Xiong. Preprint, 2026. [Code]

Research Experience

SparcAI Inc. Jan 2026 - Present
Research Collaboration and Research Intern | Advisor: Dr. Yufei Wang Remote

- **Channel Match.** Distilled TRELLIS.2 to two steps with **4.5x faster shape sampling**. Implemented a fused Triton JVP FlashAttention kernel and latent calibration for a frozen PBR texture generator.
- **Compact 4D scene states.** Built a sparse autoencoder with shared static features, temporal slots, and a 2D Gaussian surfel decoder. Stored 32-frame MultiCamVideo clips in **0.40M values on average**. With frozen per-frame features, a policy reached **81.3% StackCube success with 1,000 demonstrations**.
- **VLA manipulation.** Built robot-centric point maps and aligned RGB-D, robot state, language, and actions for Xiaomi-Robotics-1 and pi0.5. Added failure-case data augmentation and evaluated grasping, placing, and pushing on a **physical SO-101**.
- **3D data & training.** Helped curate roughly **3M 3D assets** through filtering, geometry repair, and multi-view generation. Helped run, monitor, and debug **distributed training across 128 GPUs**.

Research Experience (Continued)

Robot Skill Factory | **Open-source Research Project** 2026

Built a system that combines language-model planning with skill reuse, on-demand tracker training, and regression checks. Evaluated continuous execution across motion tracking, walking, and object interaction on a **simulated Unitree G1** using one RTX 4090.

William & Mary Sep 2024 - Present

Ph.D. Researcher | Advisor: Dr. Yanhai Xiong Williamsburg, VA

- **VTV-FM.** Co-developed second-order phase-space flow matching with minimum-acceleration Hermite bridges, analytic terminal-velocity closure, and optimal-transport pairing. Evaluated distributions, PDE fields, images, and short videos; **CIFAR-10 FID 2.99 with 100-step Velocity Verlet**.
- **Physical world models.** Studied interaction history, explicit physical factors, and geometric output constraints across **six manipulation scenes** with PointWorld/DINO-WM. In PointWorld, reduced wasted pushing by **37% on four ambiguous-mobility scenes**; completed **PointWorld-to-SO-101 transfer**.
- **HyCO.** Co-developed entropy/disagreement-triggered RL-to-diffusion solution completion for TSP, maximum independent set, and orienteering; reduced **TSP-1000 optimality gap from 15.4% to 4.6%** over POMO.
- **LENS.** Combined learned local potential fields, topology guidance, and bounded Conflict-Based Search for multi-agent planning. Matched an 85M-parameter MAPF model's OOD performance with **less than 0.2% of its training data** in the global-map protocol; separately evaluated online unknown-map navigation.
- **TAIR** / **Track, Then Bind.** Bound object identity at visible evidence and maintained it using SAM2.1 tracks with frozen Qwen3-VL-32B. Reached **48.6% versus 37.1% on 1,845 held-out TOC-Bench questions** and 100% on both synthetic VET-Bench tasks; tested real videos and matched-budget controls.
- **UP-Bench.** Built an open-source HoloOcean 3D navigation benchmark with currents, obstacles, and partial observability; integrated classical planners and SAC/hierarchical policies with automated success, collision, path-quality, control-effort, and compute evaluation.

Earlier Research & Industry

Cambricon Technologies Co., Ltd. | **Digital Logic Engineer** Aug 2022 - Jul 2024

Built functional and system-level performance verification for AI accelerator chips, including low-latency data flow and error-handling paths.

National University of Singapore | **Master's Research Project** Feb 2021 - Nov 2021

Designed spoof surface plasmonic structures in CST Microwave Studio and built a **2.4 GHz USRP B210** measurement platform. Used MATLAB/LSTM pulse reconstruction for heart/respiratory-rate estimation and DNN abnormal-signal classification.

Technical Skills & Practical Experience

Software: Python, PyTorch, Triton, Hugging Face, DDP, vLLM, sparse convolution, gsplat, Apptainer, Linux.

Robotics: Isaac Sim/Lab, ManiSkill, MuJoCo, OmniGibson, HoloOcean, SO-101; SAC, MPPI/CEM; RGB-D/video pipelines and scene asset/physics configuration.

Deployment: assembled and wired a 3D-printed SO-101, diagnosed intermittent cable faults, and improved VLA data collection with fixed lighting.

Teaching, Service & Honors

Teaching Assistant, William & Mary: DATA 305, Problem Solving with Generative AI (Spring 2026); DATA 303, Introduction to Data Visualization (Fall 2025).

Reviewer: AAAI (2025, 2026).

Summer Research Fellowship, William & Mary (2025, 2026).

Scholarship for Model Student, University of Electronic Science and Technology of China (2018, 2019).